NAIVE BAYES

Deadline 23/3/2021

Obligatory

Conditional independent

$$P(x_1, x_2 \mid Y) = P(x_1 \mid x_2, Y) \cdot P(x_2 \mid Y)$$
$$= P(x_1 \mid Y) \cdot P(x_2 \mid Y)$$

More generally

$$P(x_1, \dots x_d \mid Y) = \prod_j P(x_j \mid Y)$$

NB

$$\underbrace{P(y \mid x_1, \dots x_d)}_{\text{class posterior}} \propto \underbrace{P(y)}_{\text{prior}} \cdot \underbrace{\prod_j P(x_j \mid y)}_{\text{data likelihood}}$$

Find y which maximizes the prob in the posterior

$$y_{NB} = \arg\max_i P(y_i) \prod_j P(x_j \mid y_i)$$

Learning NB (TRN)

For each class y_i

Estimate $P(y_i)$ on TRN. ← prior.

$P(y_i) = \frac{n_i}{n}$, n_i: # ex ∈ class y_i

n: # ex ∈ TRN.

For each attribut x_j

For each value v_{jk} of x_j

estimate $\boxed{P(x_j = v_{jk} | y_i)}$ on TRN

Classify NB

For each example x of test data

For each class y_i

estimate $P(y_i | x) = P(y_i) \cdot \prod_j P(x_j | y_i)$

Return y_i with maximal $P(y_i | x)$

$P(x | y)$

1) continuous variables

Assume Normal dist.

TRN : Estimate + save.

cond mean

cond. std.

TST : Estimate prob

using Normal density fun.

$$N(x_j | \mu_{j|y}, \sigma^2_{j|y}) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu_{j|y})^2}{2\sigma^2_{j|y}}\right)$$

2) Discrete variables

TRN : relative frequences

$$P(X_j = v_{jk} \mid y_i) = \frac{n_{jk}}{n_i}, \quad n_{jk} = \#\text{ex} \in y_i \text{ where } x_j = v_{jk}$$

n_i : # ex $\in$ class y_i

$$P(y \mid x) \propto P(y) \cdot \prod_j P(x_j \mid y)$$

$$P(y=1 \mid x) = P(y=1) \cdot \prod_j P(x_j \mid y=1)$$

$$P(y=0 \mid x) = P(y=0) \cdot \prod_j P(x_j \mid y=0)$$